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Thermodynamic and economic analysis of a hydrogen production process from medical waste by plasma gasification

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ABSTRACT

To reduce the environmental pollution caused by pathogens and radioactive substances carried by medical waste, and alleviate the shortage of non-renewable energy, this work proposes a novel overall process for hydrogen production from medical waste by plasma gasification coupled with ionic liquid-based CO_2 capture. The validity of the model is verified by the experimental data. Based on the simulation results, the effects of medical waste types, different gasification agents and carbon conversion rate on the syngas compositions, H_2/CO ratio, carbon conversion rate, higher heating value and exergy efficiency are explored. The results show that the rate of hydrogen production from surgical masks is more than twice that of the other two types of medical waste. Steam is the best gasification agent. At this time, the higher heating value of syngas is 26.72 MJ/kg and the exergy efficiency is 78.25%. A comprehensive analysis of the thermodynamic efficiency and technical economy of the overall process shows that the total exergy efficiency is 70.84%. The production cost is 2115.74 USD and the raw material consumption is 2.61 t/t H_2. This study can provide a promising theoretical guidance for the resource treatment of medical waste.

1. Introduction

Adequate energy is one of the important factors to ensure people's normal life and social production activities. The excessive consumption of fossil energy has led to excessive greenhouse gas emissions and aggravated global climate change (Dincer and Aydin, 2023; Wang et al., 2019). As a clean energy, H_2 can be a good substitute for traditional energy and serve as an energy carrier (Longden et al., 2022; Tang et al., 2023). The physical and chemical properties of clean energy enable them to have cleaner combustion structures and compositions than fossil energy. These energies produce fewer pollutants during combustion and can slow the growth and accumulation of CO_2 (Manigandan et al., 2023).

Medical waste (MW) is infectious, pathological, cytotoxic and radioactive waste, which will not to degrade naturally (Li et al., 2020). With the sound development of the world's medical and health system, the recognition of disposable medical products has increased, resulting in an accelerated growth rate of medical waste. Since the Coronavirus disease 2019 (COVID-19) outbreak, 129 billion surgical masks and 65 billion gloves have been used globally every month (Chen et al., 2021). The annual production of MW in China are 2.244 million tons, of which surgical masks are about 130 thousand tons. The waste generated in the medical field has caused great concern. It is a general trend to explore a harmless treatment technology that can transform MW into fuel.

The most primitive approach to deal with MW is to decompose it into harmless substances through landfills (Bolan et al., 2023). Ozbay et al. (2021) found that non-compliant landfills could lead to the infiltration of toxic substances from MW, which could have adverse effects on life and environment. World Health Organization (WHO) recommended on-site high temperature treatment prior to disposal of these wastes to reduce the risk of infection during transport and storage. According to Kampf et al. (2020) and Chin et al. (2020), thermal disinfection was considered to be an effective method of sterilizing contaminants including the COVID-19 virus. Saknimit et al. (1988) found that heat treatment at 80 °C for 1 min reduced viral infectivity by at least 4log10. The above studies have shown that thermochemical conversion technology at high temperature can completely eliminate bacteria or pathogens carried by MW (Voudrias, 2016).

Gasification is the process of converting hydrocarbons into a mixture

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Nomenclature

MW	Medical waste.
COVID-19	Coronavirus disease 2019.
MWTH	MW to produce hydrogen by plasma gasification.
MWPG	MW plasma gasification.
WGS	Water-gas shift.
CC	CO_2 capture.
GD	Gas dehydration.
SGM	Surgical mask.
SGMTH	Surgical mask to hydrogen by plasma gasification.

Symbols

$\dot{E}^{ph}$	Physical exergy.
$\dot{E}^{ch}$	Chemical exergy.
$\dot{E}_D$	Exergy loss rate.
ε	Exergy efficiency.
η	Raw material consumption.
TCI	Total capital investment.
WC	Working capital.
FCI	Fixed capital investment.
EI	Equipment investment.
PC	Production cost.

of H_2, CO, CO_2, CH_4 and long-chain hydrocarbon (Purnomo et al., 2021). Compared with other heat treatment technologies, gasification technology has the advantage of flexibility of feedstock and it can process various types of waste without significantly changing the product composition (Raj et al., 2023). These disadvantages are that furans, dioxins and a large amount of particulate matter will be produced during the gasification process, which seriously affects the ecological environment and human health (Erdogan and Yilmazoglu, 2021). Taking proper measures to deal with MW is essential to avoid public health problems and environmental pollution.

Plasma gasification is an effective and little environmental pollution gasification technology in the treatment and energy recovery of solid wastes such as plastic wastes (Galaly and Dawood, 2023; Soni et al., 2021), waste tires (Ma et al., 2022) and biomass (Gao et al., 2021). The technology converts the organic part of the solid raw materials into HHV syngas for subsequent fuel production and power generation systems. The inorganic part can be processed into ash. More importantly, no toxic substances were detected in the product. MW is composed of hydrocarbons with high volatile matter and carbon content. The heating value of MW is much higher than that of coal and biomass (Miranda et al., 2013).

The Gases Institute of the National Academy of Sciences of Ukraine and Chernets et al. (2010) built an experimental device for plasma pyrolysis for the treatment of MW. The experimental results showed that the treatment of 1 kg MW required 1 kW•h of power and produced 1.1–1.4 nm^3 syngas. After heat evolution calculation, the obtained syngas can form an energy carrier with a capacity of 3.59–4.56 kW·h for the production of alternative fuels. Messerle et al. (2018) studied the process of plasma gasification of biomedical waste through a combination of experiments and numerical simulations. In addition, the comparison of energy balances with plasma gasification and conventional incineration showed that plasma technology had higher energy efficiency. Singh et al. (2022) studied the production of hydrogen-rich syngas by steam injection of a plasma gasifier through modeling and response surface methods, and determined the optimal operating range of the gasifier. Tavares et al. (2019) evaluated the effect of the equivalence ratio of the plasma gasification behavior based on a previously developed Aspen Plus model for plasma gasification of municipal solid waste. The results showed that the lower heat value of the syngas obtained when ER= 1 was higher when oxygen was used as the gasification agent and the temperature was 1500 °C. From the above studies, it is evident that using plasma gasification technology to treat medical waste can effectively recover chemical energy.

Another advantage of plasma gasification technology is the ability to produce syngas and other valuable products. Among them, syngas is often used to produce a series of chemicals such as clean fuel hydrogen and methanol. Inayat et al. (2021) established the heat integral model of jujube seed steam gasification process and heat integration process through mixed integer nonlinear programming in MATLAB. The results showed that the H_2 in the product increased from 60 mol% to 70 mol% as the temperature increased from 600 °C to 700 °C. The main reason was that coke gasification reaction and water gas shift reaction were linearly related to temperature. Al-Qadri et al. (2022) used Aspen Plus to simulate the case of converting plastics into hydrogen fuels based on gasification and steam methane reforming and evaluated the energy efficiency, greenhouse gas emissions, and technical economics of the process. The hydrogen production rate of this process is about 52.2%, and the energy efficiency is 68.37%. Babatabar et al. (2021) established a hydrogen production process model of the integrated configuration of biomass gasification and water-gas shift (WGS) reaction in Aspen Plus. The effects of reaction temperature and steam volume in WGS reactor on CO conversion and H_2/CO ratio were studied. According to the thermodynamic equilibrium prediction results, the conversion rate of CO and the ratio of H_2/CO in the inlet stream of WGS reactor decreased with the increase of WGS reaction temperature, and increased with the increase of steam volume.

Table 1 compares the differences between this work and previous studies. The research on the overall process of plasma gasification from MW and subsequent hydrogen production is limited. And the hydrogen production process of plasma gasification coupled with ionic liquid-based CO_2 capture has not been investigated yet. Therefore, a novel overall process of plasma gasification of MW using steam as gasification agent to produce crude syngas and further produce H_2 is proposed in this work. Based on the simulation results, this work explores the effects of the composition of medical waste and the type of gasification agent on performance parameters such as H_2/CO ratio and heat value of syngas. The feasibility of the process is proved based on the analysis of thermodynamic efficiency and technical economy. The proposal of this process will provide a way of thinking for medical waste treatment and energy shortage.

2. Model Description and Validation

In this work, the process of MW to produce hydrogen by plasma gasification (MWTH) is simulated by Aspen Plus V11. The process of

Table 1
Comparison of present and previous studies.

Feedstock	System type	Exergy efficiency	Refs.
Coal	Gasification + chemical looping system	47.22%	Yan et al. (2022)
Coal	Partial gasification & combustor + power generation + CO_2 capture system	55.68%	Ye et al. (2020)
Rice straw	Air separation + gasification + WGS + pressure swing absorption system	67.00%	Bhattacharya et al. (2014)
Medical waste	Plasma gasification + solid oxide fuel cell + steam turbine cycle + autothermal reforming system	63.20%	Zhao et al. (2022)
Cassava	Steam reforming + WGS system	72.61%	Madeira et al. (2020)
Medical waste	Plasma gasification + WGS + ionic liquid-based CO_2 capture + gas dehydration system	70.84%	This work

MWTH mainly includes four basic units: MW plasma gasification unit (MWPG), water-gas shift unit (WGS), CO_2 capture unit (CC) and gas dehydration unit (GD). Fig. 1 shows the flowsheet of MWTH. The physical properties, main models and critical parameters of units of the process are shown in Table 2.

2.1. MW plasma gasification process

Plasma gasification technology to treat solid waste refers to adding the solid waste to the plasma gasifier after preliminary treatment. Air, N_2, Ar or other gasification agents are ionized under high pressure to generate a high temperature of 1650–11600 °C. Decompose organic solid waste into H_2, CO, CH_4 and other combustible gases. Non-combustible solid waste into vitrified solid waste and other harmless residues (Longden et al., 2022). At the same time, the incorporation of the arc into the plasma significantly reduces the emission of environmentally harmful gases, such as furans and dioxins, into the atmosphere (Cai and Du, 2021). Plasma gasification is a technology that enables the harmless treatment of MW containing various bacteria and viruses and the production of gaseous fuels. MW can be divided into cellulose products, plastic products protein products and so on. And the gauze, surgical masks (SGM) and absorbable gut we chose belong to cellulose products, plastic products and protein products in this study, respectively. In this study, we select SGM, gauze and absorbable catgut as typical medical waste samples. Table 3 provides the proximate analysis and ultimate analysis and HHV of the three types of MW samples of the SGM (Brillard et al., 2021), gauze and absorbable catgut (Deng et al., 2014), where the HHV of the MW samples is calculated by Eq.(1) (Arteaga-Pérez et al., 2013).

$$HHV_{Feedstocks} = 0.3491 \times C + 1.1783 \times H - 0.1034 \times O - 0.0151 \times N - 0.0211 \times Ash \quad (1)$$

The synthesis gas composition is estimated based on the Gibbs free energy minimization method. As a consequence, this study adopts the RYield and RGibbs modules as the plasma gasification model for MWTH. The process simulation of plasma gasification of MW is shown in Fig. 2.

The process simulation of plasma gasification of MW is divided into four parts: fuel preheating, pyrolysis, drying and gasification. The MW is preheated by the HEATER module and then passes into the plasma gasifier. The energy required in this process is provided by the Q2 heat flow of the H2 module, which maintained the energy balance of the reaction. The DECOMP module simulates the pyrolysis of an unconventional solid (MW) to its constituent elements C, H, O, N, S at higher temperatures. Q1 can transfer the decomposition heat of the pyrolysis process to the gasification module to maintain the energy balance of the MW gasification process. After the raw material is pyrolyzed, it enters the SEP1 module in the form of moisture, C, H_2, O_2 and ash to remove moisture. The moisture to be removed is calculated by the calculation module according to the proximate analysis and ultimate analysis of MW. SYNGAS and gasification agent PLAS-GAS react in the HTGZ high temperature gasification reaction zone at about 2500 °C to ensure the complete decomposition of the feedstock. The PLASMA module is used to simulate the direct current non-transfer plasma torches. The gasification agent is rapidly heated to 4000 ℃ and produces a plasma flow before entering the HTGZ. This high temperature change only exists inside the plasma arc and the nearby temperature will drop rapidly. A typical plasma torch efficiency is considered to be 86% (Janajreh et al., 2013). After the SYNGAS1 containing syngas and ash is cooled by the H2 module, it enters the SEP2 separation module as SYNGAS2 and the ash is completely removed in the form of liquid slag to obtain SYNGAS3. The product SYNGAS3 enters the LTGZ low-temperature gasification reaction zone to undergo further reaction and is mixed with the steam separated before gasification, and finally discharges in the form of crude syngas.

In previous work, Qi et al. (2021) have completed the validation of the plasma gasification model. In order to prove the accuracy of the

Table 2
Property methods, main models and critical parameters of operation units.

Unit	Property method	Main models	Key parameters
MWPG	PR-BM	RYield and RGibbs	m(MW)= 13 t/h; P = 1.01325 bar
WGS	RK-SOAVE	RPlug	Mole ratio H_2/CO= 2.15; P = 36 bar
CC	NRTL-RK	RadFrac	T = 25 °C; P = 1.01325 bar
GD	NRTL	Sep	Split fraction of 0 (H_2O), 1(H_2), 0.02(CO), 0.02(CH_4); P = 5 bar

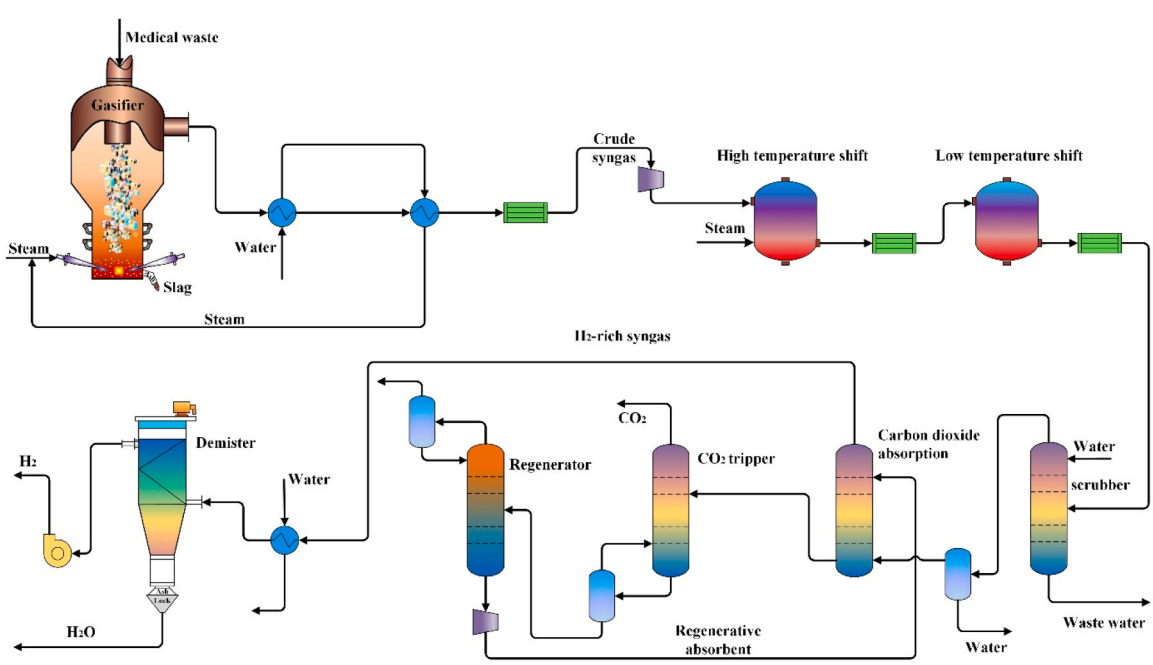

Fig. 1. Flowsheet of medical waste to produce hydrogen by plasma gasification.

Table 3
Ultimate analysis and Proximate analysis of medical wastes (dry basis).

Sample	Ultimate analysis (wt%)					Proximate analysis (wt%)				HHV[a] (MJ/kg)
	C	H	O	N	S	Moisture	VM	FC	Ash	
Surgical mask (Brillard et al., 2021)	84.17	14.93	0.70	n.d.	n.d.	5.00	92.90	1.90	0.20	46.90
Gauze (Deng et al., 2014)	42.15	6.35	50.93	n.d.	0.13	5.79	90.91	3.22	0.08	17.06
Absorbable catgut (Deng et al., 2014)	43.16	7.60	33.25	13.59	0.8	13.33	76.59	8.48	1.60	20.38

n.d. below detection.
[a] on raw basis.

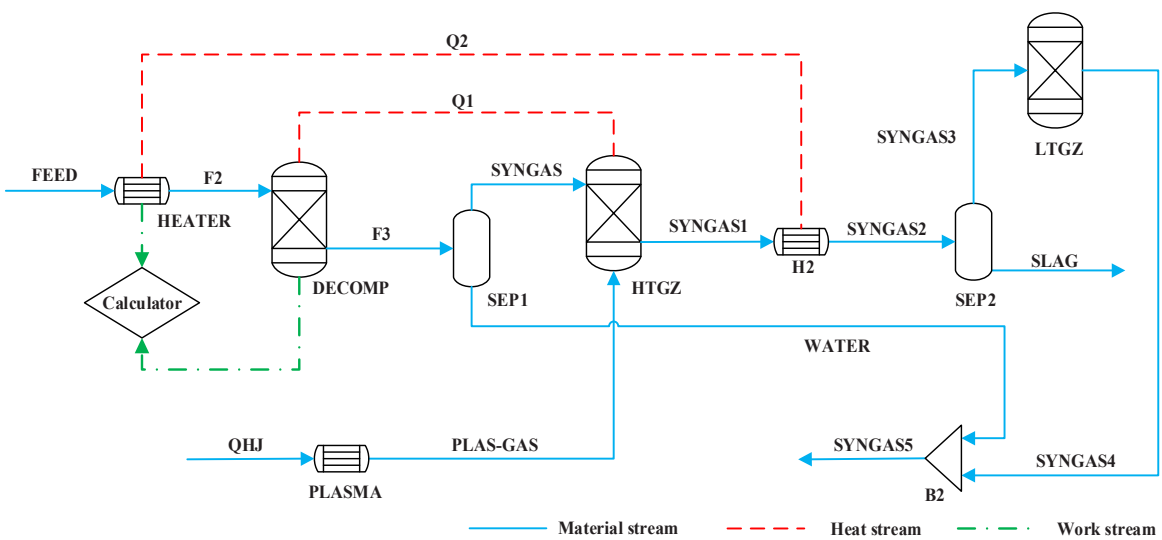

Fig. 2. Plasma gasification process modeling of medical waste.

model again, the simulation results of the model for H_2 production by plasma gasification of peach core are compared with the experimental data of Diaz et al. (2015) The model validation results are shown in Table 4. The CO content obtained from the simulation results based on plasma gasification of peach core is 7.3% different from the experimental results. This is because Diaz et al. did not clearly indicate the experimental temperature for plasma gasification of peach core. We assumed complete conversion of C during the simulated plasma gasification of peach core. With the continuous increase of plasma gasification temperature, more C will be converted into gaseous products. And usually when the temperature is above 1000 °C, the thermal stability of CO is better than that of CO_2. Therefore, the simulation results of the CO content will be greater than the experimental results, and the simulation results of the PG model are within a reasonable range. Therefore, this model can accurately simulate and calculate the composition of syngas produced by plasma gasification of MW.

The compositions of crude syngas obtained by plasma gasification of the three MW samples with air as the gasification agent is shown in Table 5 and Fig. 3. It can be seen that under the action of air gasification agent, the gaseous products of MW after plasma gasification mainly include six components: H_2O, H_2, CO, CO_2, CH_4 and H_2S. Among them, the ratio of H_2 and CO is the largest. That is, the gaseous product of plasma gasification MW is mainly composed of syngas. By comparison, it is found that the H_2 produced from SGM was the most, accounting for about 71.13 mol% of the total. It is more than 1.5 times that of the other two MWs samples. Combined with the ultimate analysis of MW in Fig. 3, the H element content of SGM is 2.35 times of the H element content of gauze and 1.96 times of the H element content of absorbable catgut. Using gauze as raw material produced the most CO, which is 54.17 mol %. The H_2O produced by absorbable catgut as raw material is the highest, 9.42 mol%. This is because among the three samples, the content of oxygen element and Moisture are the highest in gauze and absorbable catgut, respectively. In summary, it can be concluded that the composition content of the gas products obtained by MW plasma gasification has a linear relationship with the element content.

Table 4
Validation of the plasma gasification model (mole fraction (%)).

Gas composition	Simulation	Experiment (Diaz et al., 2015)
H_2	53.6	57.0
CO	26.1	18.8
CO_2	15.6	15.7
CH_4	4.7	4.6

Table 5
Three kinds of medical wastes plasma gasification of crude syngas compositions (mole fraction (%)).

Composition	Surgical mask	Gauze	Absorbable catgut
H_2O	2.38	7.61	9.42
H_2	71.13	36.86	45.85
CO	26.47	54.17	44.31
CO_2	6.36 E-04	1.30	0.12
CH_4	0.02	2.20E-05	2.05E-04
H_2S	0	0.06	0.30

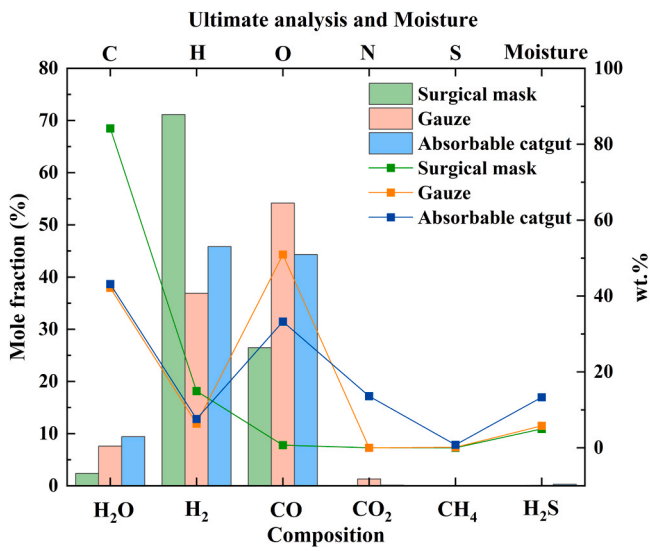

Fig. 3. The relationship between the composition of gaseous products and Ultimate analysis and Moisture.

2.2. Hydrogen production process

It can be seen from Table 3 that the mass fraction of element C in the SGM is 84.17%, and its content is much higher than other elements. In addition, the temperature of the SGM for plasma gasification is above 1650 °C, which makes the crude syngas obtained from the plasma gasification process with strong thermal stability and high content of CO. In this work, the WGS is used to convert CO into H_2, and the involves reaction formula is shown in Eq.(2) (Zhou et al., 2023).

$$CO + H_2O \leftrightarrow CO_2 + H_2 \Delta H(298K) = -41.1kJ \bullet mol^{-1} \tag{2}$$

In this work, the WGS is divided into two stages (high temperature shift 350–500 °C and low temperature shift 180 °C~300 °C) to improve CO conversion rate and H_2 yield. A model of a WGS is established using the RPlug module based on a rigorous kinetic model. The physical property method is RK-SOAVE, and the reaction kinetic equation is mentioned in Eq.(S1-S3). Similarly, in order to prove the accuracy of the WGS model and the reference kinetic parameters, industrial data (Zhong, 2016) of water gas shift hydrogen production were used in the model. Table 6 shows the comparison of model simulation results with industrial data. The simulation results of the WGS model are evident within a reasonable range. Therefore, the model can accurately simulate and calculate the process of CO conversion into H_2.

If high purity H_2 is to be obtained, measures must be taken to remove the CO_2 and water contained in the syngas removed from the WGS. The commonly methods use CO_2 capture in the industry mainly include hot potash absorption process based on chemical absorption, organic amine absorption process and low temperature methanol washing process based on physical absorption, propylene carbonate absorption process and ionic liquid absorption process. Among them, physical absorption method uses physical principles to absorb CO_2 without chemical reaction. Absorbent recycling requires less regenerative energy. Ionic liquid is a new type of green solvent with low volatility, good thermal stability, non-flammability and adjustable structure, which has great potential in the field of CO_2 capture. In 1999, Blanchard et al. (1999) observed that the solubility of CO_2 in 1-butyl-3-methylimidazolium hexafluorophosphate ([BMIM][PF_6]) could reach 0.6 mol fraction at 8 MPa, and ionic liquid did not dissolve in CO_2. On this basis, the solubility of CO_2 in other five ionic liquids was studied in depth. Table 7 lists the CO_2 absorption capacity of six ionic liquids. As a typical example, the molar solubility of CO_2 in 1-butyl-3-methylimidazolium tetrafluoroborate ([BMIM] [BF_4]) can be as high as 0.444 at 39.7 bar and 323.15 K, which was comparable to that of CO_2 in MEA solution. Ma et al. (2017) simulated a CO_2 capture process based on novel ionic liquids with a 25% reduction in energy consumption compared to MEA-based processes. The results showed that ionic liquid-based CO_2 capture process was more competitive than the conventional MEA-based CO_2 capture process. It is considered that the cost of [OMIM][PF_6] is about three times that of [BMIM][PF_6]. Therefore, [BMIM][PF_6] is selected as the absorbent in this work. NRTL-RK is used as a physical property method to simulate a model to CC and purify syngas (Zhang, 2018).

The H_2-rich syngas removed from the CC enters the GD. The GD mainly includes two processes of gas cooling and demisting (Li, 2013). It utilizes the principle that the saturated vapor pressure of water in the syngas decreases with the decrease of the gas temperature. First, the water-containing syngas is fed into the condenser and cooled to a temperature at which the water vapor condenses into droplets. Then through the mist separator to separate the droplet and gas to realize the drying of the syngas. High purity H_2 can be obtained after CO_2 capture and gas dehydration.

3. Methods

In order to demonstrate the potential advantages of MWTH, thermodynamic analysis and economic benefit evaluation are performed using the SGM to produce hydrogen by plasma gasification (SGMTH) as an example in this work.

3.1. Thermodynamic analysis of SGMTH

Exergy analysis is often used to reveal the irreversibility and thermodynamic efficiency of a system. Exergy can be used to measure the part of a process that maximizes the conversion of energy into useful work. For steady-state systems, the exergy of streams mainly includes physical exergy ($\dot{E}^{ph}$) and chemical exergy ($\dot{E}^{ch}$). Eq.(3) (Hasanzadeh et al., 2023) is the computational formula of exergy. When the SGM is used as solid raw material, only chemical exergy is considered. The chemical exergy of the dry SGM is estimated by Eq.(4) (Qi et al., 2023).

$$\dot{E} = \dot{E}^{ph} + \dot{E}^{ch} \tag{3}$$

$$\begin{aligned}\dot{E}_{SGM} =& 363.439\omega(C) + 1075.633\omega(H) - 86.308\omega(O) + 4.417\omega(N) \\ &+ 190.788\omega(S) - 21.1\omega(Ash)\end{aligned} \tag{4}$$

where $\dot{E}^{ph}$ can be calculated by Eq.(5) (Hasanzadeh et al., 2023). $\dot{E}_{SGM}$ is the chemical exergy of surgical mask. $\omega(C)$, $\omega(H)$, $\omega(O)$, $\omega(N)$, $\omega(S)$ and $\omega(Ash)$ are the mass fractions of C, H, O, N, S, Ash of the dry surgical mask, respectively. $\dot{E}^{ch}$ can be calculated by Eq.(6) (Hasanzadeh et al.,

Table 6
Validation of the water-gas shift model (mole fraction (%)).

Gas composition	Simulation	Industrial data (Zhong, 2016)
H_2	39.8	40.9
CO	13.5	12.4
CO_2	32.3	33.6
CH_4	11.4	11.5
H_2S	0.14	0.14

Table 7
Absorption capacities of ionic liquids for CO_2 (Blanchard et al., 2001).

Ionic liquid	Pressure (bar)	Temperature (K)	Mole CO_2 / Mole IL
[BMIM] [PF_6]	30.07	323.15	0.513
[BMIM] [BF_4]	39.70	323.15	0.444
[BMIM] [NO_3]	30.08	323.15	0.357
[N-bupy] [BF_4]	41.95	323.15	0.441
[EMIM] [$EtSO_4$]	42.49	323.15	0.287
[OMIM] [PF_6]	44.27	323.15	0.789

2023).

$$\dot{E}^{ph} = \dot{m}(h_i - h_0) - T_0(s_i - s_0) \tag{5}$$

$$\dot{E}^{ch} = \sum z_i e_i \tag{6}$$

In addition, the $\dot{E}^{ch}$ of the mixed gas can be calculated as (Hasanzadeh et al., 2023):

$$\dot{E}^{ch} = RT_0 \sum z_i ln z_i + \sum z_i \dot{E}^{ch}_{0,i} \tag{7}$$

where R is the universal gas constant, and its value is 8.314 kJ/kmol/K. The standard chemical exergy ($\dot{E}^{ch}_{0,i}$) of each substance is shown in Table S1.

The exergy loss rate is equal to the difference between the input and output exergy values (Yu et al., 2022). It can be calculated by Eq.(8).

$$\dot{E}_D = \sum \dot{E}_{in} - \dot{E}_{out} = \sum \dot{E}_{in,stream} - \dot{E}_{out,stream} + \sum \dot{Q}\left(1 - \frac{T_0}{T}\right) + \sum \dot{W} \tag{8}$$

The exergy efficiency (ε) is used as the index for exergy analysis (Yu et al., 2022). Calculated as follows:

$$\varepsilon = \frac{\sum \dot{E}_{out}}{\sum \dot{E}_{in}} = 1 - \frac{\dot{E}_D}{\sum \dot{E}_{in}} \tag{9}$$

3.2. Economic benefit evaluation of SGMTH

Process economic benefit evaluation is a significant part of chemical process design. It is a decisive standard to measure whether a process is feasible. Moreover, the production cost of the process plays a crucial role in the economic benefit of the process, among which the consumption cost of raw materials accounts for a large proportion. The ratio of the SGMTH can indicate the consumption of raw materials, calculated by Eq.(10) (Qi et al., 2023):

$$\eta = \frac{m_{SGM}}{m_{H_2}} \tag{10}$$

In Eq.(10), m_{SGM} is the quality of the surgical mask consumed, and m_{H_2} is the quality of the product hydrogen.

Working capital (WC) and fixed capital investment (FCI) constitute total capital investment (TCI). The function of WC is to keep the factory running normally, it is made up of parts such as finished products and taxes. Expenses such as purchasing equipment and pipes belong to the FCI. The TCI is normally determined by the cost of the equipment, and the TCI is estimated using Eq.(11) (Xu et al., 2023) according to the corresponding percentages of the other components of the TCI in the cost of the equipment. Table S2 lists the components of TCI and their corresponding percentages.

$$TCI = C_E(1 + \sum RF_i) \tag{11}$$

where C_E is the equipment investment of the chemical process, RF is the proportional factor and i is the component of the investment.

The cost of purchasing equipment, tools and appliances is called equipment investment (EI). It is a necessary aspect of capital investment in chemical processes. The calculation is based on Eq.(12) (Xu et al., 2023) and the fundamental data on EI and proportional factor in Table S3.

$$EI = C_E\left(\frac{Q}{Q_B}\right)^M f_M f_P f_T \tag{12}$$

where C_E is the equipment investment, C_B is the reference project equipment, Q_B is the specification of the reference process, and Q is the specification of the research process. M represents the specification exponent. f_M represents the correction factor for building materials, f_p represents the design pressure correction factor, and f_T represents the design temperature correction factor.

Production cost (PC) is another important index for analyzing the economic performance of a chemical processes. PC is the sum of raw material costs, utility costs, worker wages and so on, as shown in Eq.(13). The assumptions for calculating PC are shown in Table S4. Furthermore, the depreciation cost is calculated using the straight-line method, assuming a 15-year useful life and a 4% residual value. The cost of the remainder of the PC is calculated based on their percentage of the PC (Xu et al., 2023).

$$PC = C_R + C_U + C_{O\&M} + C_D + C_{AC} + C_{DSC} \tag{13}$$

4. Results and discussion

In this work, in order to explore the best gasification agent, thermodynamic efficiency and economic benefit evaluation of SGMTH, the process simulation results of SGMTH are analyzed and discussed.

4.1. Gasification agent selection

The crude syngas obtained from the MWPG mainly depends on the types of raw material and gasification agent (Sharma and Sheth, 2016). Kuo et al. (2020) investigated the effect of three different gasification agents on the plasma gasification performance of different carbonized biomass. The study showed that the syngas obtained when steam and CO_2 were used as gasification agents has low sulfur content. The syngas obtained from carbonized pine as raw material was of low impurity and high quality. Therefore, this study selects the SGM as a typical example to study the effects of O_2/N_2 mixed gasification agent (the ratios of O_2 are 21%, 35%, 45%, 55%, 65%, 75%, 95%, 100%) and O_2/Steam mixed gasification agent (the ratios of O_2 are 0%, 20%, 40%, 60%, 80%, 100%) on crude syngas compositions, the ratio of H_2/CO, carbon conversion rate, HHV of syngas and the exergy efficiency in PG model. The composition of crude syngas and the ratio of H_2/CO are shown in Table 8.

As shown in Figs. 4 and 5, the effects of the mixed gasification agent O_2/N_2 and O_2/Steam on the performance indexes of the model are investigated in this work. Using O_2/N_2 mixed gasification agent for plasma gasification, the carbon conversion rate, the HHV of the syngas and the exergy efficiency to increase with the continuous increase of the O_2 ratio. Carbon conversion increases from 0.40 to 0.79. The resulting syngas heat value of the unit is 21.71 MJ/kg and the exergy efficiency is 0.66. However, the H_2/CO ratio decreases by 1.34 and the production capacity of hydrogen is significantly reduced. When using O_2/Steam mixed gasification agent, with the increase of O_2 ratio, all the indicators show a downward trend. The analysis results show that in the plasma

Table 8
Components of crude syngas from plasma gasification of the surgical mask using different mixed gasification agents (mole fraction (%)).

Component	H_2	CO	CO_2	CH_4
O_2/N_2 mixed gasification agent				
21%	45.82	17.06	4.10E-04	0.01
35%	50.39	25.20	8.83E-04	0.01
45%	51.82	29.27	1.23E-03	0.01
55%	53.04	32.65	1.58E-03	0.02
65%	54.09	35.50	1.93E-03	0.02
75%	55.02	37.92	2.28E-03	0.02
95%	56.71	41.84	3.04E-03	0.02
100%	57.30	42.58	3.53E-03	0.02
O_2/Steam mixed gasification agent				
0%	67.44	31.87	0.68	8.20E-05
20%	63.53	35.93	0.53	1.21E-04
40%	60.47	39.18	0.35	2.01E-04
60%	57.98	41.87	0.14	5.07E-04
80%	56.71	43.27	3.58E-03	0.02

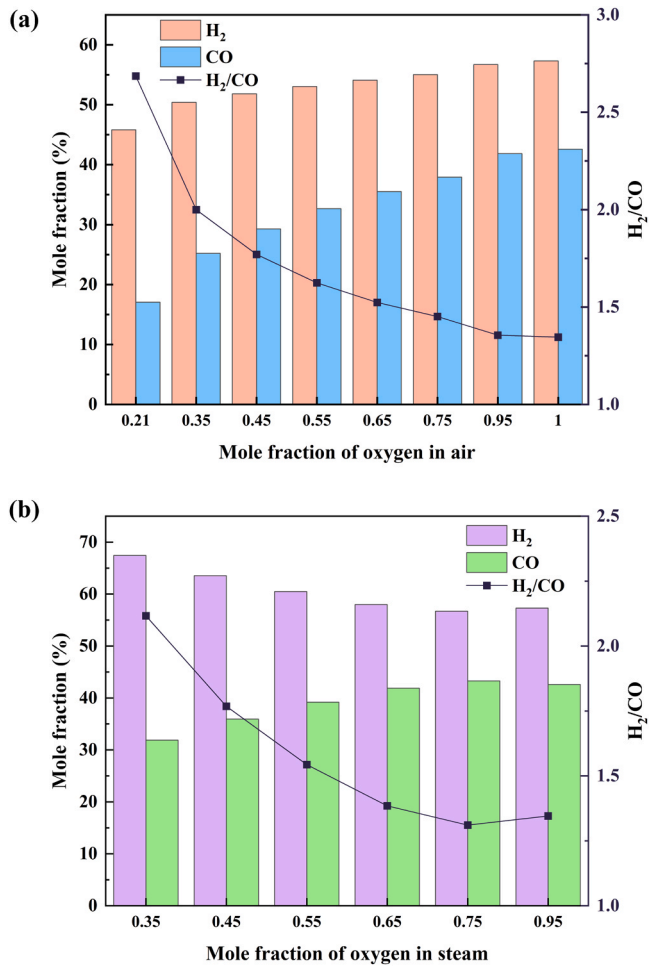

Fig. 4. (a) The effects of O_2/N_2 mixed gasification agent on the composition of syngas and H_2/CO ratio. (b) The effects of O_2/Steam mixed gasification agent on the composition of syngas and H_2/CO ratio.

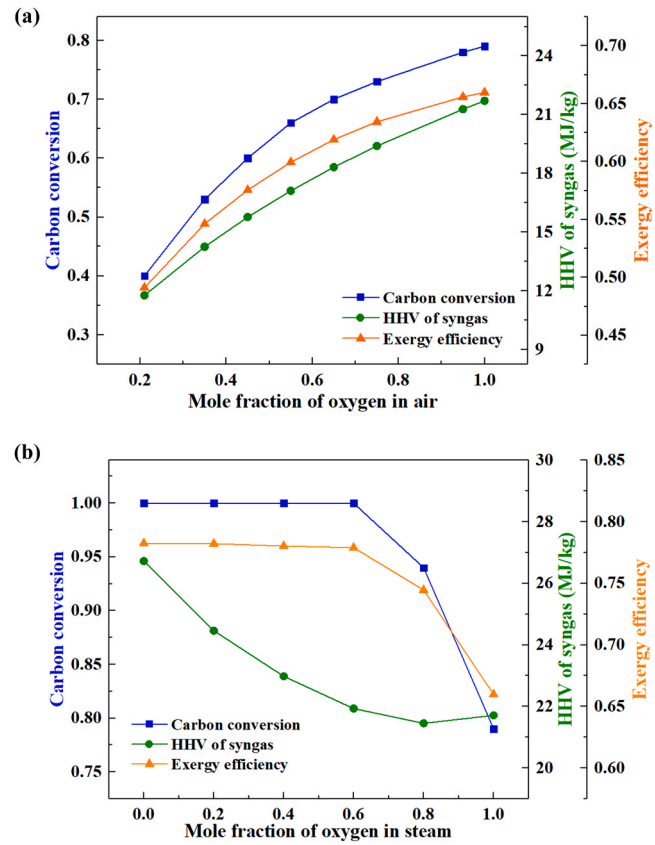

Fig. 5. (a) Effect of O_2/N_2 mixed gasification agent on process performances. (b) Effect of O_2/Steam mixed gasification agent on process performances.

gasification model, the continuous injection of oxygen into the gasification agent will inhibit the production capacity of H_2 and increase the production rate of CO. It is concluded that the performance of the model is the best when steam is used as the gasification agent of the plasma gasification model of the SGM. At this time, the H_2/CO ratio and HHV of the syngas are 2.12 and 26.72 MJ/kg respectively, and the exergy efficiency of the plasma gasification process is as high as 0.78.

4.2. Thermodynamic efficiency analysis

Exergy analysis is more accurate and reliable than previous energy analysis. The main reason is that it takes into account the quantity and quality of energy. The SGMTH process is discussed through the results of process simulation and exergy calculation. Table 9 lists the results of the simulated SGMTH process.

Figs. 6 and 7 respectively show the exergy flow in the whole process and the exergy loss ratio of each unit. Exergy of MW as the main raw material to produce syngas is 168.22 MW. Exergy of MWPG is 78.97 MW. Exergy of crude syngas derived from plasma gasification of MW is 181.24 MW, while exergy of steam is 12.19 MW. The exergy efficiency of plasma gasification is 78.25%. Exergy loss of the unit is mainly due to high temperature caused by plasma, high power consumption and irreversibility of plasma gasification of MW into syngas.

Exergy efficiency of WGS unit is 81.58%. 226.82 MW is the total exergy, of which the cold energy required by WGS is 18.12 MW. The exergy loss of the unit is 39.84 MW. By dividing the crude syngas at the inlet into two streams and adjusting the amount of steam, the purpose of controlling the temperature of the reactor and improving the conversion rate of CO is achieved. At the same time, it has a positive effect on reducing the exergy loss caused by the high irreversibility of water-gas shift reaction. The Exergy of CC unit is 199.96 MW. Exergy output includes 179.06 MW of hydrogen-rich syngas, 5.44 MW of CO_2 and 15.46 MW of exergy loss. Exergy efficiency is 92.27%, and exergy loss is mainly caused by viscous flow of ionic liquid. The GD unit has an input exergy of 187.73 MW. Exergy output includes purified syngas exergy of 175.94 MW, impurity exergy of 0.02 MW and exergy loss of 11.77 MW. Exergy efficiency is 93.72%. Exergy loss is mainly caused by irreversibility of heat transfer and phase transformation of water.

Therefore, according to the above analysis, the sum of exergy input in SGMTH is 316.35 MW. The exergy efficiency of the whole SGMTH process is 70.84%. The exergy loss of the four units is 92.25 MW. Among them, the SGM plasma gasification process is the unit with the largest exergy loss, accounting for about 44.3% of the total exergy loss. The hydrogen drying process is the unit with the lowest exergy loss, accounting for about 9.8% of the total exergy loss.

4.3. Economic benefit evaluation

In this work, the economic benefits of SGMTH are evaluated by analyzing raw material consumption, TCI and PC. According to Eq.(10), the raw material consumption of SGMTH is 2.61 t/t H_2.

After calculation according to Eq.(12) and Table 9, the EI is 14.34×10^6 USD. According to Eq.(11), TCI of SGMTH is 54.96×10^6 USD. According to Eq.(13), The PC of SGMTH is 2115.74 USD/t. By comparing the contribution of each component in PC in Fig. 8, it is found that raw materials and utilities are the main contributors to the

Table 9
Simulation results of the surgical mask to hydrogen by plasma gasification.

Parameter	Unit	MW	Crude syngas	Shifted syngas	Rich H_2 gas	H_2
Temperature	°C	25	450	307	406	50
Pressure	bar	1	36	36	36	5
Mass flows	kg/h	13000	28148.32	44607.80	6567.18	5345.47
Mole flows	kmol/h		2658.43	3572.41	2707.28	2640.95
H_2O	kmol/h		0.16	67.59	67.57	1.35
H_2	kmol/h		1792.81	2639.60	2639.6	2639.6
CO	kmol/h		847.36	0.55	-	-
CO_2	kmol/h		18.10	864.66	-	-
[BMIM][PF_6]	kmol/h		-	-	0.1	-
Sum exergy	MW	168.22	181.24	185.05	179.06	175.94

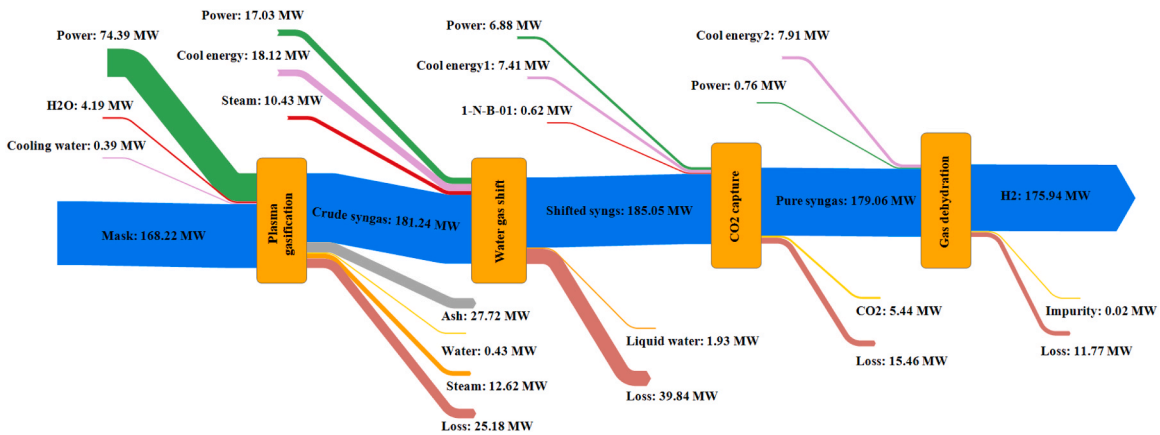

Fig. 6. Exergy flow of the surgical mask to hydrogen by plasma gasification.

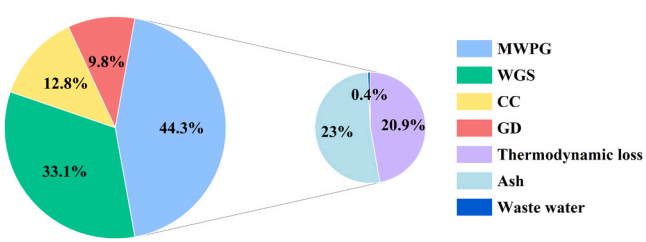

Fig. 7. Exergy loss of each unit in the process.

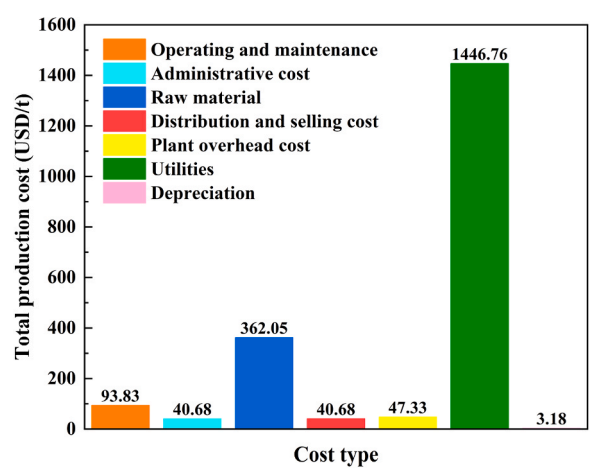

Fig. 8. Contribution of each component in production cost.

production cost of this process, accounting for 18.54% and 74.08% of PC, respectively. The utility consumption is mainly due to the high electricity consumption of the plasma gasifier.

5. Conclusion

This study aims to propose a novel hydrogen production process using MW as raw material, and plasma gasification coupled with ionic liquid-based CO_2 capture to achieve efficient and harmless treatment of MW. The effects of MW composition, carbon conversion and gasification agent type on process performance are explored. The thermodynamic efficiency and technical economy of the overall process of SGMTH are evaluated. The specific results are as follows:

1. The mole fraction of H_2 obtained from plasma gasification of SGM is 71.13%, which is 34.27% and 25.28 higher than that of gauze and absorbable gut, respectively. Each component content of the gaseous product is linearly related to the element content of MW.
2. The effects of mixed gasification agents O_2/N_2 and O_2/Steam on process performance indexes are investigated. It is concluded that steam is the best gasification agent. At this time, the H_2/CO ratio and HHV of the syngas are 2.12 and 26.72 MJ/kg respectively. The exergy efficiency of the plasma gasification process is as high as 78.25%.
3. The total exergy efficiency of SGMTH is 70.84%. The SGM plasma gasification process is the unit with the largest exergy loss, accounting for about 44.3% of the total exergy loss. The hydrogen drying process is the unit with the lowest exergy loss, which is 11.79 MW.
4. The raw material consumption of SGMTH is 2.61 t/t H_2. The TCI is 54.96×10^6 USD. The PC is 2115.74 USD. In the six categories of expenses, the consumption of utilities is the most expensive, followed by raw material collection and pretreatment.

In summary, the MWTH process can not only realize the timely disposal of waste MW to reduce public health problems and

environmental pollution, but also convert MW into clean energy to effectively alleviate the energy shortage problem. we will consider exploring a more efficient CO_2 capture process and conducting a comprehensive life cycle assessment of the plasma gasification to hydrogen production process by medical waste in future work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Notes

The authors declare no competing financial interest.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.psep.2023.08.007.

References

Al-Qadri, A.A., Ahmed, U., Jameel, A.G.A., Zahid, U., Usman, M., Ahmad, N., 2022. Simulation and Modelling of Hydrogen Production from Waste Plastics: Technoeconomic Analysis. Polymers 14, 2056.
Arteaga-Pérez, L.E., Casas-Ledón, Y., Pérez-Bermúdez, R., Peralta, L.M., Dewulf, J., Prins, W., 2013. Energy and exergy analysis of a sugar cane bagasse gasifier integrated to a solid oxide fuel cell based on a quasi-equilibrium approach. Chem. Eng. J. 228, 1121–1132.
Babatabar, M.A., Saidi, M., 2021. Hydrogen production via integrated configuration of steam gasification process of biomass and water-gas shift reaction: Process simulation and optimization. Int. J. Energy Res. 1–17.
Bhattacharya, A., Das, A., Datta, A., 2014. Exergy based performance analysis of hydrogen production from rice straw using oxygen blown gasification. Energy 69, 525–533.
Blanchard, L.A., Dan, H.C., Beckman, E.J., Brennecke, J.F., 1999. Green processing using ionic liquids and CO_2. Nature 399 (6731), 28–29.
Blanchard, L.A., Gu, Z.Y., Brennecke, J.F., 2001. High-Pressure Phase Behavior of Ionic Liquid/CO_2 Systems. J. Phys. Chem. B. 105 (12), 2437–3444.
Bolan, S., Padhye, L.P., Kumar, M., Antoniadis, V., Sridharan, S., Tang, Y.Y., Singh, N., Hewawasam, C., Vithanage, M., Singh, L., Rinklebe, J., Song, H., Siddique, K.H.M., Kirkham, M.B., Wang, H.L., Bolan, N., 2023. Review on distribution, fate, and management of potentially toxic elements in incinerated medical wastes. Environ. Pollut. 321, 121080.
Brillard, A., Kehrli, D., Douguet, O., Gautier, K., Tschamber, V., Bueno, M.A., Brilhac, J. F., 2021. Pyrolysis and combustion of community masks: Thermogravimetric analyses, characterizations, gaseous emissions, and kinetic modeling. Fuel 306, 121644.
Cai, X.W., Du, C.M., 2021. Thermal Plasma Treatment of Medical Waste. Plasma Chem. Plasma P. 41 (1), 1–46.
Chen, C., Chen, J.A., Fang, R., Ye, F., Yang, Z.L., Wang, Z., Shi, F., Tan, W.F., 2021. What medical waste management system may cope With COVID-19 pandemic: Lessons from Wuhan. Resour. Conserv. Recy. 170, 105600.
Chernets, O.V., Korzhyk, V.M., Marynsky, G.S., Petrov, S.V., Zhovtyansky, V.A., 2010. Electric arc steam plasma conversion of medicine waste and carbon containing materials. Proc. 2008 17th Int. Conf. Gas. Disch. Their Appl.
Chin, W.H., Chu, T.S., Perera, R.A., Hui, P.Y., Yen, H.L., Chan, C.W., 2020. Stability of SARS-CoV-2 in different environmental conditions. Lancet Microbe 5247 (20), 30003.
Deng, N., Cui, W.Q., Wang, W.W., Zhang, Q., Zhang, Y.F., Ma, H.T., 2014. Experimental study on co-pyrolysis characteristics of typical medical waste compositions. J. Cent. South Univ. 21 (12), 4613–4622.
Diaz, G., Sharma, N., Leal-Quiros, E., Munoz-Hernandez, A., 2015. Enhanced hydrogen production using steam plasma processing of biomass: Experimental apparatus and procedure. Int. J. Hydrog. Energ. 40 (5), 2091–2098.
Dincer, I., Aydin, M.I., 2023. New paradigms in sustainable energy systems with hydrogen. Energy Convers. Manag. 283, 116950.
Erdogan, A.A., Yilmazoglu, M.Z., 2021. Plasma gasification of the medical waste. Int. J. Hydrog. Energ. 46 (57), 29108–29125.
Galaly, A.R., Dawood, N., 2023. Energy Recovery and Economic Evaluation for Industrial Fuel from Plastic Waste. Polymers 15 (11), 2433.
Gao, N.B., Mwenya, H.M., Cui, Q., Li, R.D., 2021. Critical assessment of plasma tar reforming during biomass gasification: A review on advancement in plasma technology. J. Hazard. Mater. 421, 126764.
Hasanzadeh, R., Mojaver, P., Azdast, T., Khalilarya, S., Chitsaz, A., Rosen, M.A., 2023. Decision analysis for plastic waste gasification considering energy, exergy, and environmental criteria using TOPSIS and grey relational analysis. Process Saf. Environ. Prot. 174, 414–423.
Inayat, A., Shahbaz, M., Khan, Z., Inayat, M., Mofijur, M., Ahmed, S.F., Ghenai, C., Ahmad, A.A., 2021. Heat integration modeling of hydrogen production from date seeds via steam gasification. Int. J. Hydrog. Energ. 46, 30592–30605.
Janajreh, I., Raza, S.S., Valmundsson, A.S., 2013. Plasma gasification process: Modeling, simulation and comparison with conventional air gasification. Energ. Convers. Manag. 65, 801–809.
Kampf, G., Scheithauer, S., Lemmen, S., Saliou, P., Suchomel, M., 2020. COVID-19-associated shortage of alcohol-based hand rubs, face masks, medical gloves, and gowns: proposal for a risk-adapted approach to ensure patient and healthcare worker safety. J. Hosp. Infect. 105 (3), 424–427.
Kuo, P.C., Illathukandy, B., Wu, W., Chang, J.S., 2020. Plasma gasification performances of various raw and torrefied biomass materials using different gasifying agents. Bioresour. Technol. 314, 123740.
Li, L., 2013. Introduction and application of common gas dehydration and drying process. Technol. Dev. Chem. Ind. 42 (7), 38–41.
Li, X.Y., Zhao, H.T., Zhang, M.J., Lv, G.F., Dou, X.X., 2020. Research on the technology of plasma treatment of medical waste. Chem. Manag. 31, 63–64.
Longden, T., Beck, F.J., Jotzo, F., Andrews, R., Prasad, M., 2022. Clean' hydrogen? – Comparing the emissions and costs of fossil fuel versus renewable electricity based hydrogen. Appl. Energ. 306, 118145.
Ma, T., Wang, J.X., Du, Z.Z., Abdeltawab, A.A., Al-Enizi, A.M., Chen, X.C., Yu, G.R., 2017. A process simulation study of CO_2 capture by ionic liquids. Int. Jk. Greenh. Gas. Con. 58, 223–231.
Ma, Y.X., Qi, H.Q., Zhang, J.F., Cui, P.Z., Zhu, Z.Y., Wang, Y.L., 2022. Thermodynamic analysis of a carbon capture hydrogen production process for end-of-life tires using plasma gasification. J. Clean. Prod. 384, 135662.
Madeira, J.G.F., Oliveira, E.M., Springer, M.V., Conceição, L.T., Delgado, A.R.S., 2020. Exergy analysis of hydrogen production from manipueira biogas. Environ. Prog. Sustain. 40 (2), 13495.
Manigandan, S., Praveenkumar, T.R., Ryu, J.I., Verma, T.N., Pugazhendhi, A., 2023. Role of hydrogen on aviation sector: A review on hydrogen storage, fuel flexibility, flame stability, and emissions reduction on gas turbines engines. Fuel 352, 129064.
Messerle, V.E., Mosse, A.L., Ustimenko, A.B., 2018. Processing of biomedical waste in plasma gasifier. Waste Manag. 79, 791–799.
Miranda, M., Cabrita, I., Pinto, F., Gulyurtlu, I., 2013. Mixtures of rubber tyre and plastic wastes pyrolysis: A kinetic study. Energy 58, 270–282.
Ozbay, G., Jones, M., Gadde, M., Isah, S., Attarwala, T., 2021. Design and operation of effective landfills with minimal effects on the environment and human health. J. Environ. Public Health 2021, 1–13.
Purnomo, C.W., Kurniawan, W., Aziz, M., 2021. Technological review on thermochemical conversion of COVID-19-related medical wastes. Resour. Conserv. Recy. 167, 105429.
Qi, H.Q., Cui, P.Z., Liu, Z.Q., Xu, Z.F., Yao, D., Wang, Y.L., Zhu, Z.Y., Yang, S., 2021. Conceptual design and comprehensive analysis for novel municipal sludge gasification-based hydrogen production via plasma gasifier. Energ. Convers. Manag. 245, 114635.
Qi, H.Q., Xu, H.W., Zhang, J.F., Xu, Z.F., Zhong, L.M., Cui, P.Z., Zhu, Z.Y., Wang, Y.L., 2023. Thermodynamic and techno-economic analyses of hydrogen production from different algae biomass by plasma gasification. Int. J. Hydrog. Energy. https://doi.org/10.1016/j.ijhydene.2023.06.038.
Raj, R., Tirkey, J.V., Singh, D.K., Jena, P., 2023. Co-gasification of waste triple feed-material blends using downdraft gasifier integrated with dual fuel diesel engine: An RSM-based comparative parametric optimization. J. Energy Inst. 109, 101271.
Saknimit, M., Inatsuki, I., Sugiyama, Y., Yagami, K., 1988. Virucidal efficacy of physicochemical treatments against coronaviruses and parvoviruses of laboratory animals. Exp. Anim. 37, 341–345.
Sharma, S., Sheth, P.N., 2016. Air–steam biomass gasification: Experiments, modeling and simulation. Energ. Convers. Manag. 110, 307–318.
Singh, D.K., Tirkey, J.V., 2022. Valorization of Hazardous Medical Waste using Steam injected Plasma Gasifier: A Parametric Study on the Modeling and Multi-Objective Optimization by integrating Aspen Plus with RSM. Environ. Technol. 43 (27), 4291–4305.
Soni, V.K., Singh, G., Vijayan, B.K., Chopra, A., Kapur, G.S., Ramakumar, S.S., 2021. Thermochemical Recycling of Waste Plastics by Pyrolysis: A Review. Energ. Fuel. 35 (16), 12763–12808.
Tang, D., Tan, G.L., Li, G.W., Liang, J.G., Ahmad, S.M., Bahadur, A., Humayun, M., Ullah, H., Khan, A., Bououdina, M., 2023. State-of-the-art hydrogen generation techniques and storage methods: A critical review. J. Energy Storage 64, 107196.
Tavares, R., Ramos, A., Rouboa, A., 2019. A theoretical study on municipal solid waste plasma gasification. Waste Manag. 90, 37–45.
Voudrias, E.A., 2016. Technology selection for infectious medical waste treatment using the analytic hierarchy process. J. Air Waste Manag. 66 (7), 663–672.
Wang, Y.L., Li, G.X., Liu, Z.Q., Cui, P.Z., Zhu, Z.Y., Yang, S., 2019. Techno-economic analysis of biomass-to-hydrogen process in comparison with coal-to-hydrogen process. Energy 185, 1063–1075.

Xu, Z.F., Zhou, Y.R., Yin, K.X., Zhang, J.F., Zhu, Z.Y., Wang, Y.L., Cui, P.Z., 2023. Exergy, techno-economic and environment analysis of food waste plasma gasification and syngas chemical looping processes. J. Clean. Prod. 2023 (386), 135835.

Yan, X.S., Liu, H.L., Luo, M., Cai, J.J., Shen, R.C., 2022. Performance of hydrogen and power co-generation system based on chemical looping hydrogen generation of coal. Int. J. Hydrog. Energy 48 (30), 11180–11190.

Ye, C., Zheng, Y.Q., Xu, Y.S., Li, G.N., Dong, C., Tang, Y.J., Wang, Q.H., 2020. Energy and exergy analysis of poly-generation system of hydrogen and electricity via coal partial gasification. Comput. Chem. Eng. 141, 106911.

Yu, M.X., Liu, X.B., Chen, Y.F., Zhang, Z.H., Wang, Y.F., Zhang, J.L., Han, Y., 2022. A 4E analysis of a novel coupling process of syngas purification and CO_2 capture, transcritical CO_2 power and absorption refrigeration. Chem. Eng. J. 445, 136757.

Zhang, J.B., 2018. Design and optimization of CO_2 and H_2S separation process based on ionic liquid. Beijing University of Chemical Technology,.

Zhao, X.Y., Chen, H., Zheng, Q.W., Liu, J., Pan, P.Y., Xu, G., Zhao, Q.X., Jiang, X., 2022. Thermo-economic analysis of a novel hydrogen production system using medical waste and biogas with zero carbon emission. Energy 265, 126333.

Zhong, Huan lian xin, 2016. Environmental impact report on the Xintian Yili 20•10^8 m3/y coal-to-SNG project. Environmental Protection Inc,.

Zhou, L.M., Liu, Y.Y., Liu, S.L., Zhang, H.H., Wu, X.L., Shen, R.F., Liu, T., Gao, J., Sun, K., Li, B.J., Jiang, J.C., 2023. For more and purer hydrogen-the progress and challenges in water gas shift reaction. J. Energy Chem. 83, 363–396.